Introduction

This report details WiFi signal strength data collected and analyzed at seven different indoor locations within a university building, collected through a IndoorLocalizationA4_V2.apk Android app. With prior permission obtained from Dr. Min Hane Aung, the reserve dataset provided on Blackboard was also incorporated to supplement the training samples, ensuring a more robust analysis.

Stage 1: Data Collection Overview

This report uses lab-provided mobile phones to collect data collected at designated locations and incorporates lab-provided data to improve training samples. The referenced alternate data has also been recognized, and the code and results have been uploaded to the https://github.com/ll711/A4

Stage 2: Data Analysis and Similarity Measurement

To do 1 WiFi Signal Strength Observation

Fig. 1: Signal Data

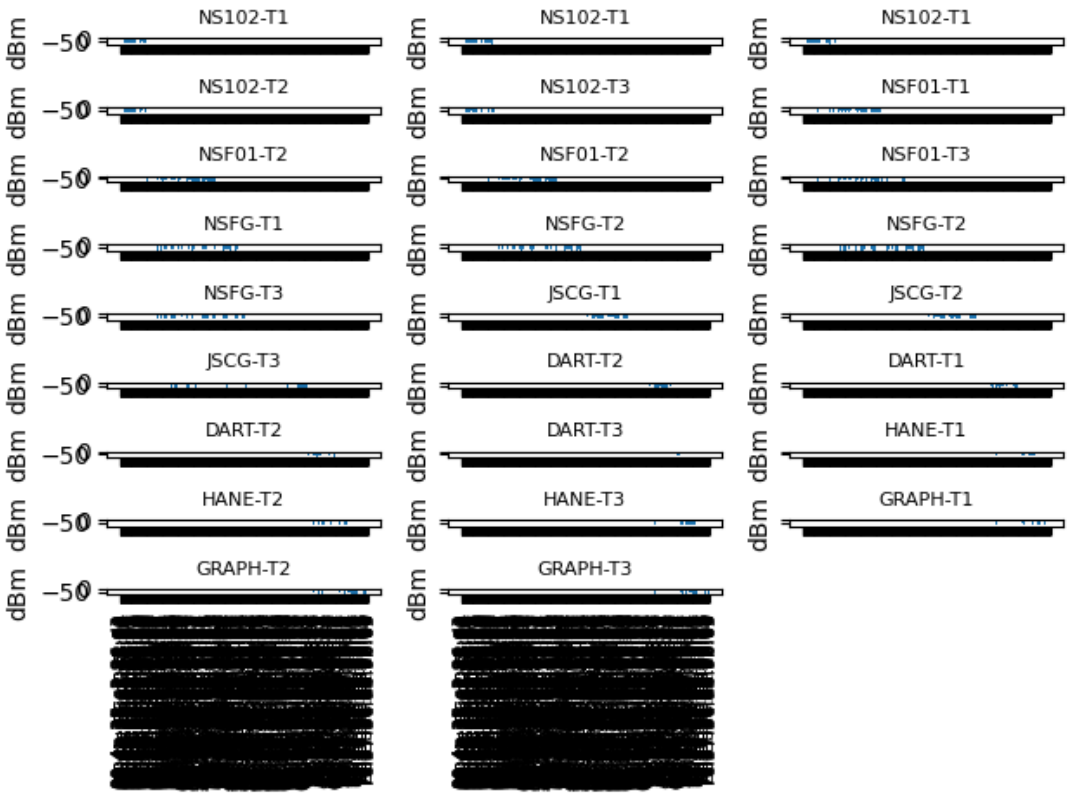


Figure 1 shows the WiFi signal strength distributions collected from the different indoor

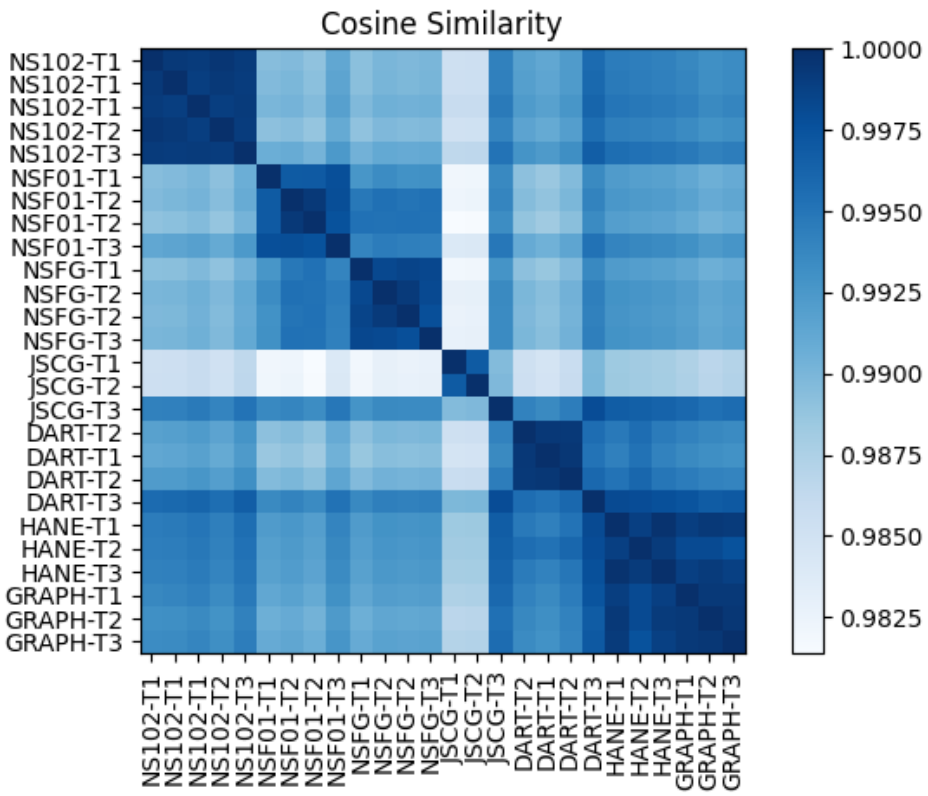
locations. Each location is represented by a high-dimensional feature vector containing signal strength values from all detected access points. In this dataset, the feature matrix consists of 26 samples and 589 access points. Because many access points were not detected at certain locations, the data are sparse and contain a large number of -100 dBm values, which represent missing or undetectable signals.

The plots indicate clear variation in signal strength across locations. Some locations, such as NS102, show comparatively stronger and more consistent signal reception, with multiple access points detected at higher signal levels. Other locations, including GRAPH and HANE, appear to have weaker overall signal strength and fewer strong detections. This suggests that the indoor layout and physical obstructions have a noticeable effect on WiFi propagation. Locations with better line-of-sight to access points tend to produce richer signal patterns, while more enclosed areas generate weaker and sparser fingerprints.

Although the signal strengths differ between locations, repeated samples from the same area remain broadly similar.

To do 2&3 Cosine Similarity Analysis

Fig. 2: Similarity Data Self



To measure the relationship between the WiFi fingerprints, cosine similarity was used to

construct a similarity matrix. This metric is appropriate for high-dimensional signal data because it compares the overall pattern of the feature vectors rather than their absolute magnitude. As required by the assignment specification, the cosine similarity matrix was computed from the WiFi signal features and 3isualized to compare the similarity between all location samples.

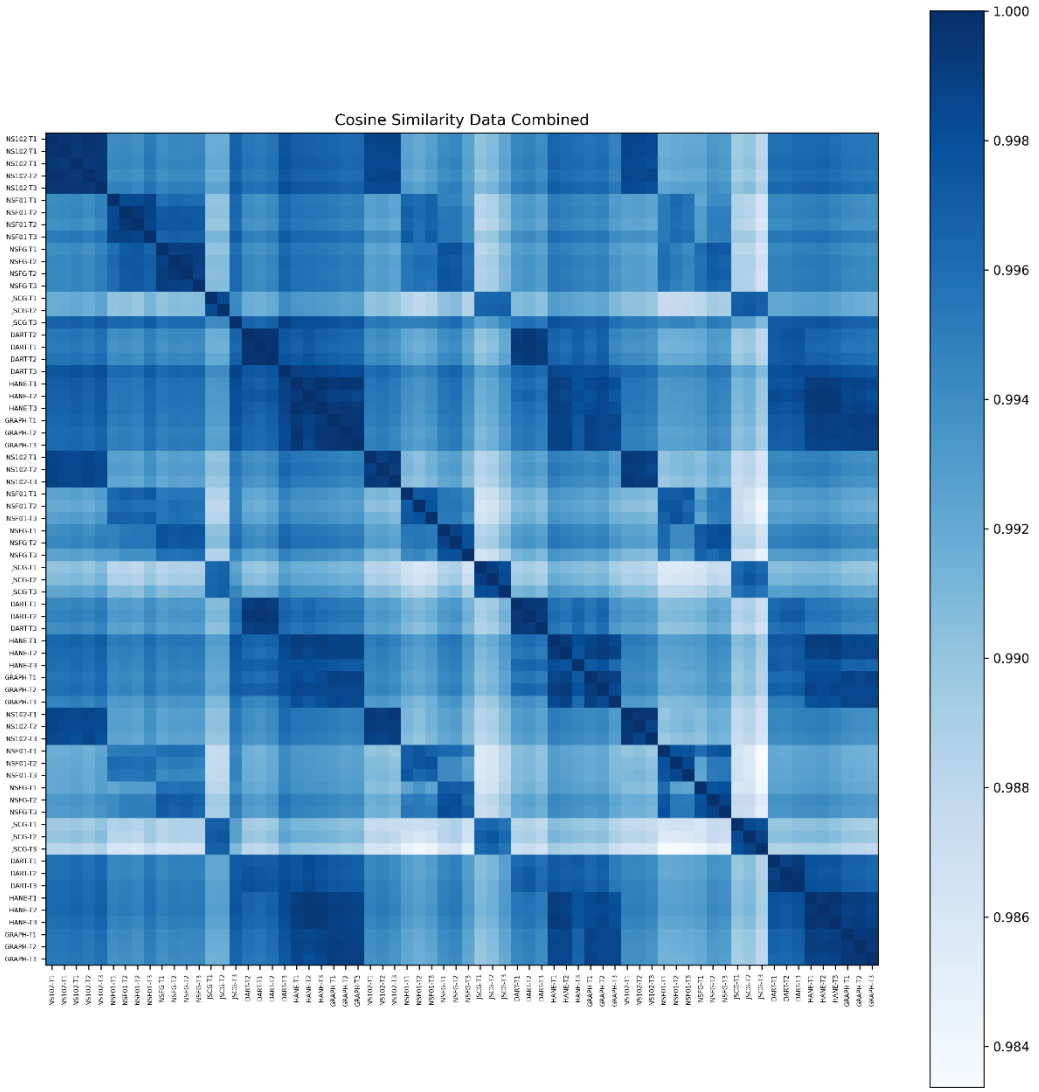
Figure 2 presents the resulting 26 × 26 cosine similarity matrix. The matrix shows a clear block structure, where samples from the same location are highly similar to one another. The diagonal values are equal to 1.0, as expected, because each sample is identical to itself. Off-diagonal values are also very high, generally remaining above 0.98, which indicates that the WiFi fingerprints across the dataset share many common signal patterns.

The strongest similarities appear between repeated scans from the same location, showing that the signal environment is relatively stable over time. NS102 in particular exhibits very high internal consistency, with its repeated samples forming a tight cluster of near-identical similarity values. By contrast, JSCG appears to be slightly more distinct from several other locations, suggesting that its signal fingerprint is more unique and therefore easier to distinguish in localization tasks.

To do 4

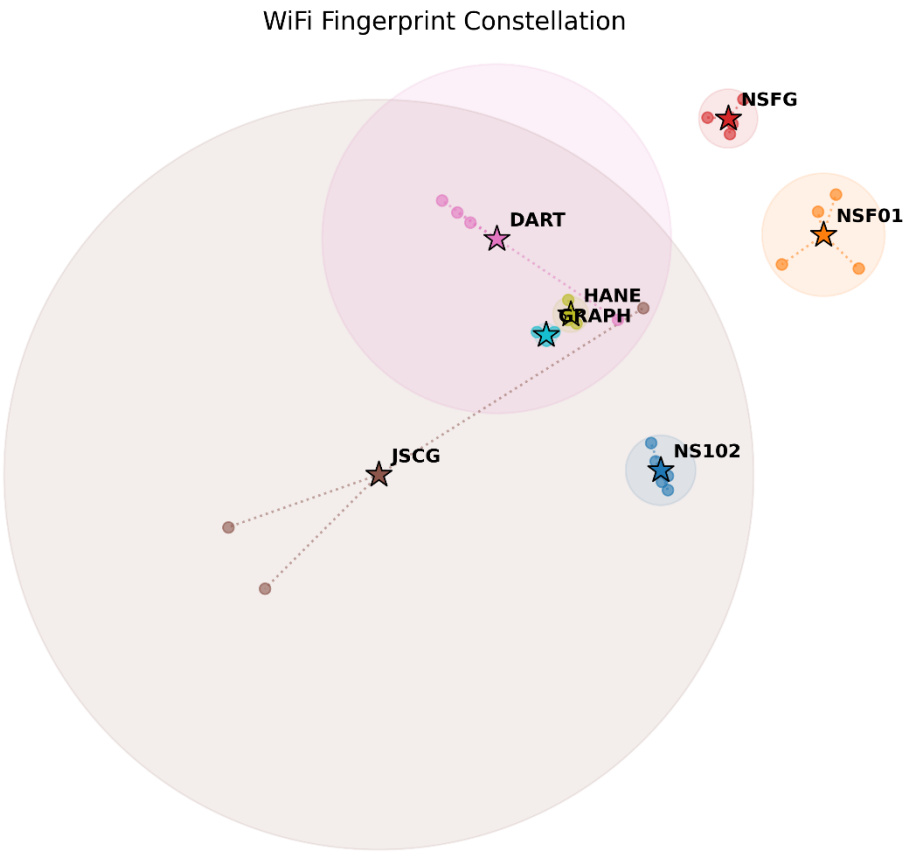
There are significant structural differences in the consistency of the date in Figure 2 and Figure 3. The heatmap for the single subject (Figure 2) exhibits large, uniform blocks of dark blue along the diagonal, indicating extremely high cosine similarity values approaching 1.0000 across different trials (T1, T2, T3) within the same task category. This suggests that the intra-subject feature distribution is highly stable and reproducible. In contrast, the combined data heatmap (Figure 3) displays a more fragmented pattern with lighter shades of blue and reduced uniformity within these blocks. The color bar scale in Figure 3 ranges from approximately 0.984 to 1.000, whereas Figure 2 is tightly compressed between 0.9975 and 1.0000, indicating a wider variance in the aggregated data. This degradation in visual uniformity in the combined plot is attributable to inter-subject variability; merging data from multiple sources introduces biological and signal noise that lowers the overall pairwise similarity compared to the highly consistent signals observed within a single individual.

Fig. 3 Similarity Data Combined:



Stage 3: User-Friendly Visualization Design

I designed a simple lo-fi visualization called a "WiFi fingerprint constellation" to make the similarity results easier to understand than a matrix of numbers. First, I took the WiFi feature vectors for each location and filled missing access point values with -100 dBm so that all samples had the same dimension. Then I used cosine similarity to measure how similar the WiFi fingerprints were, and converted this into a distance matrix using 1 - similarity, so that locations with similar patterns would appear closer together. I then applied MDS to reduce the data into two dimensions and grouped the three time samples for each location together based on their location name, such as NS102 or JSCG. In the final design, each location is shown as a central star, which represents the overall position of that place in the similarity space, while the three smaller points around it represent the repeated samples. I also added dotted lines and transparent circles to show how spread out the samples are, because a smaller cluster suggests more stable WiFi fingerprints. The axis was removed on purpose so that the figure looks more like a constellation map and is easier for a non-technical user to read at a glance.



Conclusion

This study collected and analyzed WiFi signal strength data from seven indoor locations within a university building, supplementing the measurements with a provided reserve dataset to create a feature matrix of 26 samples across 589 access points. The sparse fingerprint data revealed clear location-dependent variations, with some areas like NS102 showing strong and

consistent signals while more enclosed spaces such as GRAPH and HANE exhibited weaker reception due to physical obstructions. Cosine similarity analysis confirmed high internal consistency within the same location, with all samples from NS102 forming a tight cluster and JSCG appearing more distinct, indicating its potential for reliable localization. A user-friendly constellation visualization was developed by converting cosine similarities into distances, applying multidimensional scaling, and representing each location as a central star surrounded by its repeated sample points, thereby making the similarity structure intuitive to non-technical viewers. Overall, the WiFi fingerprints demonstrated sufficient stability and distinctiveness to support indoor localization, and the simplified visualization effectively communicated the underlying signal patterns.